

Meta-analysis Part 1

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Sources

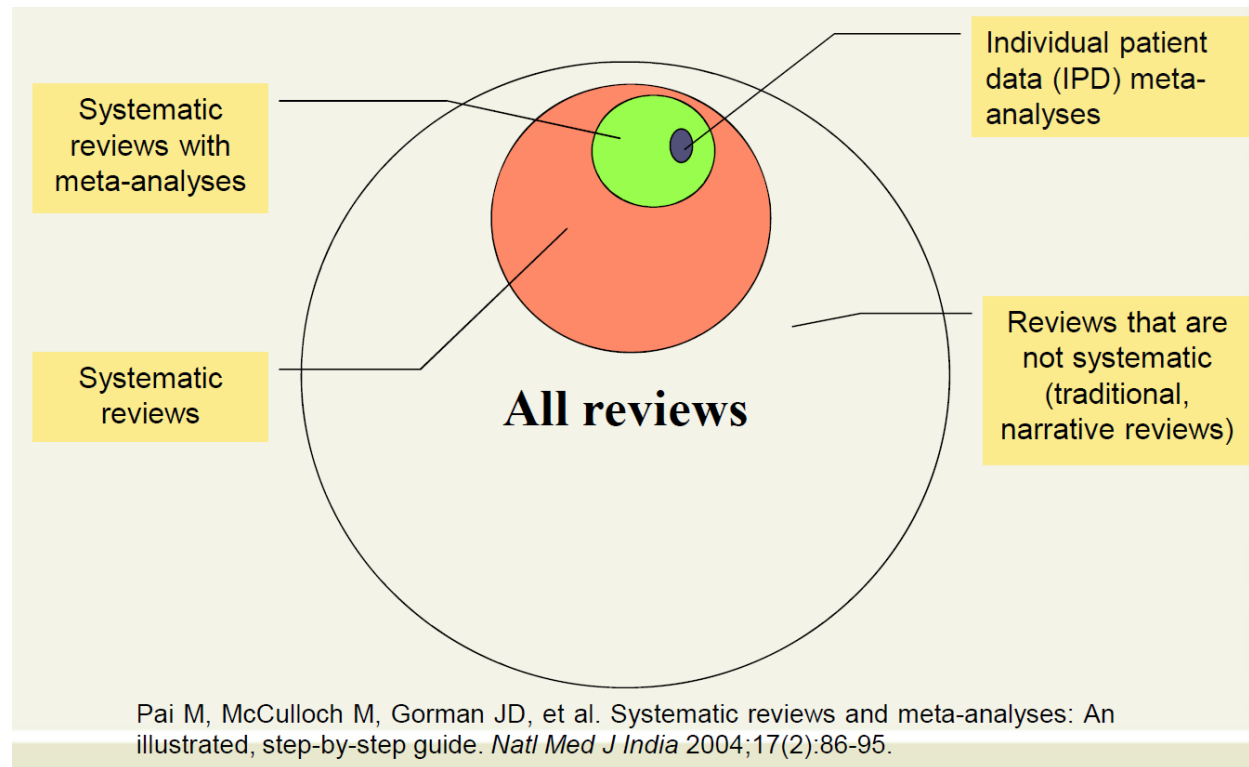
- Borenstein et al. 2009 – Introduction to Meta-analysis
- Dickersin and Li, Johns Hopkins University

Systematic Reviews

- Summarize the evidence
 - Many papers
 - Literature is large, growing, and not organized
 - Different populations, methods, quality
- Goal: evidence-based policy, practice, decision-making
- Traditional reviews:
 - No standard format
 - No clearly specified methods for identifying, selecting, and validating included information
 - Quantitative synthesis rarely used to integrate information from multiple studies

Systematic Reviews

- A review of existing knowledge that uses explicit, scientific methods
- Systematic reviews may also combine results quantitatively: meta-analysis



Systematic Reviews



Protocol:

- Research question
- Eligibility criteria
- Search
- Data abstraction
- Quality assessment
- Qualitative and/or quantitative synthesis

Systematic Reviews

- Systematic
- Transparent
- Reproducible
- Minimize bias
- Assess strength/quality of evidence
 - Note: need not only consider RCTs! But RCTs provide stronger evidence than observational studies
 - Other factors: loss to followup, blinding, ...
- Generate more precise estimates than individual studies
- Examine heterogeneity in results (why do different studies have different results?)

Components of a Meta-analysis

- Individual studies that address the “same” question
 - Pay attention to groups being compared
- Effect size: a common metric of treatment effect
- Precision of effect size
- Study weights
 - More weight to more precise studies (or some other metric of the “goodness” of a study)
- Summary effect size
 - A weighted mean
 - CI/p-value
- Heterogeneity of effect sizes

Presenting Results: Forest Plot

Impact of Statin Dose On Death and Myocardial Infarction

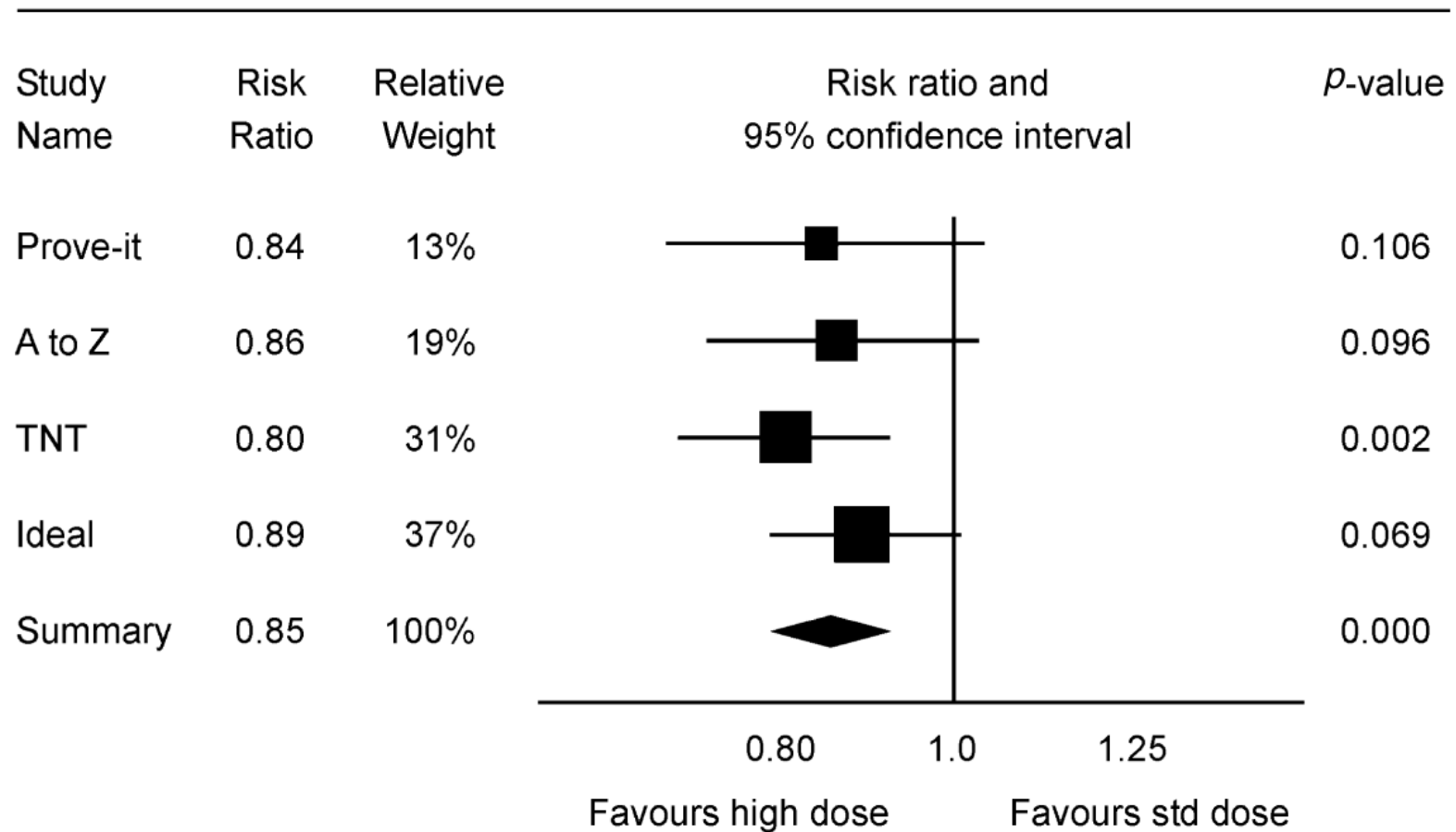


Figure 1.1 High-dose versus standard-dose of statins (adapted from Cannon *et al.*, 2006).

Choosing an Effect Size

- Effect sizes from different studies should be comparable: they measure the “same thing”
 - Example of a bad effect size: study 1 provides RR, study 2 provides RR controlling for age
- Effect sizes must be computable from the information available/obtainable
- Need to be able to extract or calculate standard error of each effect size
- Effect size must be interpretable

Choosing an Effect Size

Table 3.1 Roadmap of formulas in subsequent chapters.

Effect sizes based on means (Chapter 4)

Raw (unstandardized) mean difference (D)

Based on studies with independent groups

Based on studies with matched groups or pre-post designs

Standardized mean difference (d or g)

Based on studies with independent groups

Based on studies with matched groups or pre-post designs

Response ratios (R)

Based on studies with independent groups

Effect sizes based on binary data (Chapter 5)

Risk ratio (RR)

Based on studies with independent groups

Odds ratio (OR)

Based on studies with independent groups

Risk difference (RD)

Based on studies with independent groups

Effect sizes based on correlational data (Chapter 6)

Correlation (r)

Based on studies with one group

Mean Difference (2 Independent Samples)

$$\Delta = \mu_1 - \mu_2$$

$$D = \bar{X}_1 - \bar{X}_2$$

$$V_D = \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}$$

$$SE_D = \sqrt{V_D}$$

- Formulas will differ depending on study design: paired samples, clustered data, etc.
- Effect size has same meaning regardless of design
- For each study, compute variance/SE of the effect size appropriately

Standardized Mean Difference

$$\delta = \frac{\mu_1 - \mu_2}{\sigma}$$

- σ = population standard deviation

$$d = \frac{\bar{X}_1 - \bar{X}_2}{S_{within}}$$

- (See text for calculating SE of d)
- Provides a common metric: effect in terms of # of standard deviations

Binary Outcomes

- Options for effect size:
 - Relative risk
 - Odds ratio
 - Risk difference

Factors that Affect Precision

- Sample size (how?)
- Study design (independent samples, paired samples, clustered data, ...)
 - (which are more precise?)
- Studies that yield more precise estimates of the effect size carry more information and are assigned more weight in the meta-analysis

Fixed Effects Model

- In a fixed effects model, we assume all studies share a common true effect
- The summary effect is our estimate of this common effect size
- The null hypothesis is that this common effect is zero (for a difference) or one (for a ratio)
- All observed dispersion reflects sampling error
- Study weights are assigned with the goal of minimizing this within-study error

Fixed Effects Model

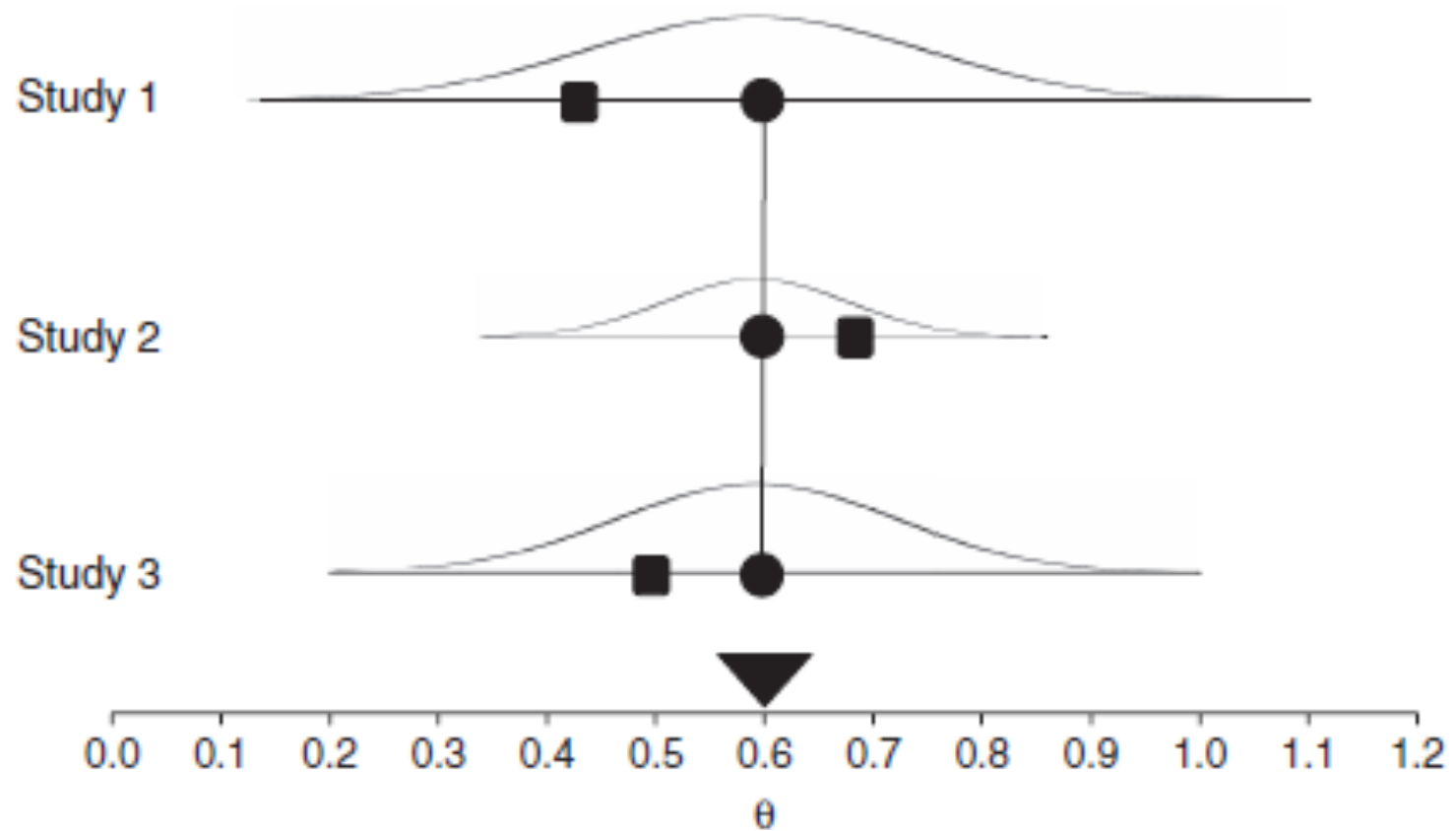


Figure 11.3 Fixed-effect model – distribution of sampling error.

Fixed Effects Model

- Weight for each study:

$$W_i = \frac{1}{V_{Y_i}}$$

- (reciprocal of the study's within-study variance)
- Summary effect size is simply the weighted mean
- SE of the weighted mean is straightforward as the studies are (should be) independent
- There are other ways of calculating the SE of the summary effect size (e.g., bootstrapping)

Random Effects Model

- In a random effects model, the true effects in the studies are assumed to have been sampled from a distribution of true effects

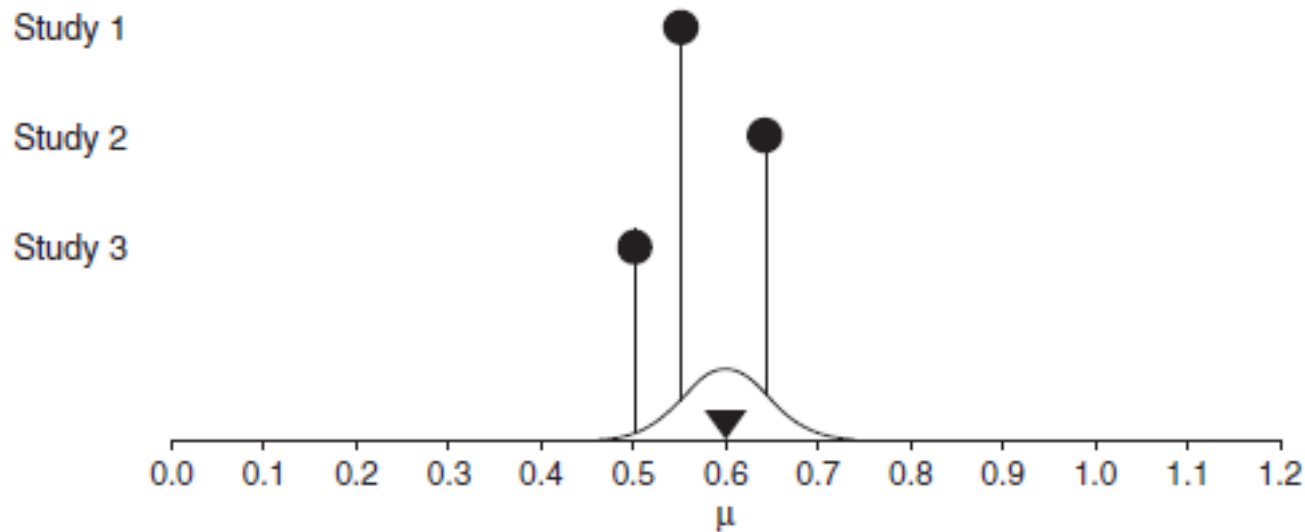


Figure 12.2 Random-effects model – true effects.

Random Effects Model

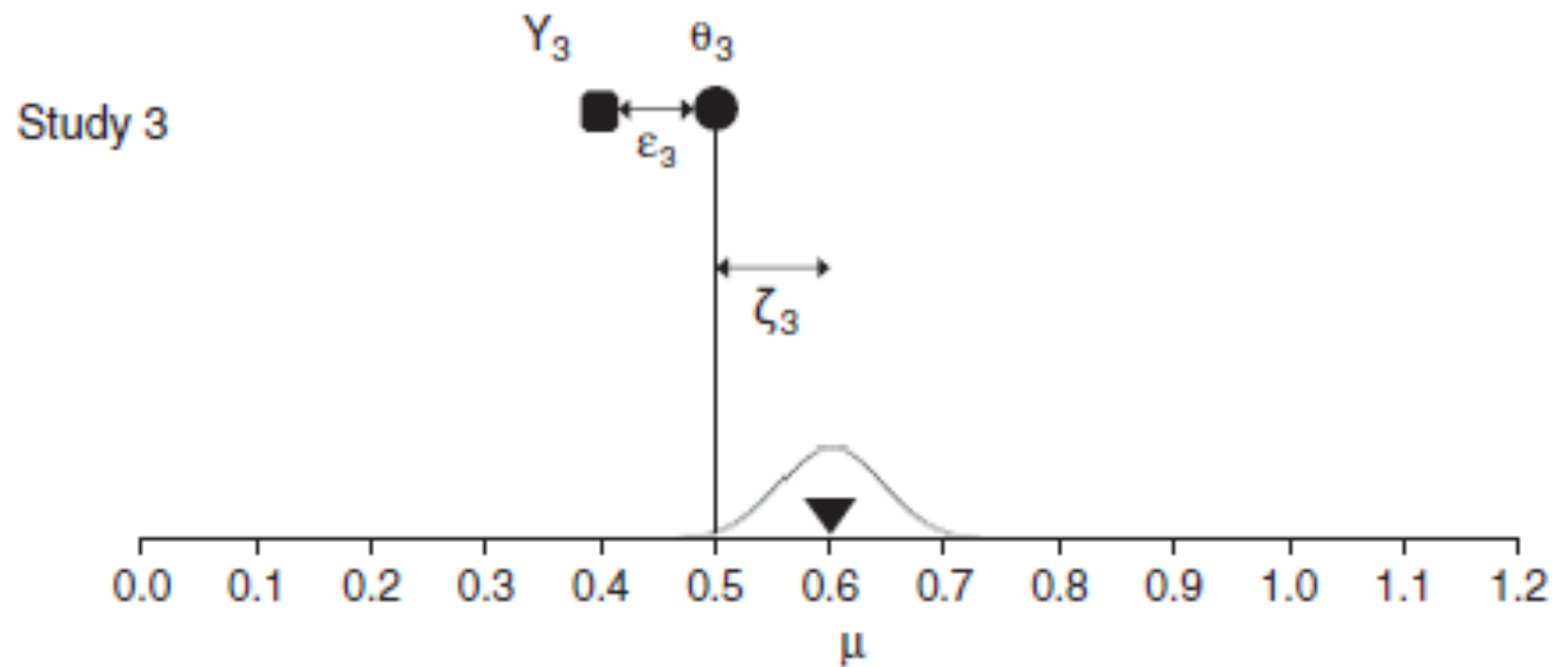


Figure 12.3 Random-effects model – true and observed effect in one study.

Random Effects Model

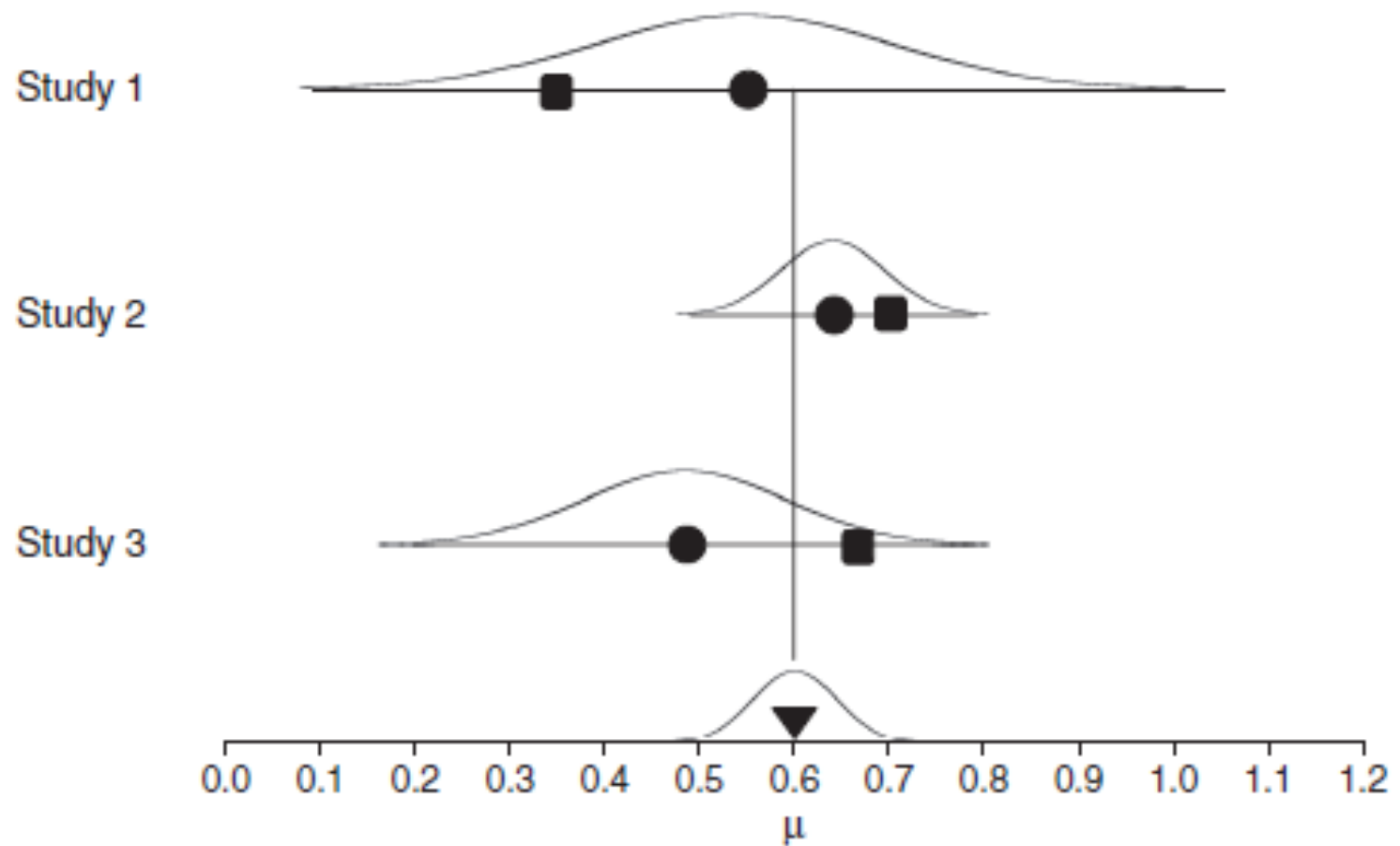


Figure 12.4 Random-effects model – between-study and within-study variance.

Random Effects Model

- The summary effect is our estimate of the mean of all true effects
- The null hypothesis is that the mean of these effects is 0.0 (equivalent to a ratio of 1.0 for ratio measures)
- We consider two sources of variation:
 - Within-study
 - Between study
- Study weights are assigned to minimize both sources of variance
- As before, the summary effect size is the weighted mean

Fixed or Random Effects Model?

- Fixed effect model estimates a single effect that is assumed to be common to every study
- Random effects model estimates the mean of a distribution of effects
- In random effects models:
 - Study weights are more balanced
 - Large studies are assigned less relative weight
 - SE and CI of the summary effect are wider

Fixed or Random Effects Model?

- Selection of a model should depend on which model makes sense (rather than looking at your data, trying to assess study heterogeneity, etc.)
- In most cases, a random effects model is more plausible

Smith et al. 2017